

Multi-affinity Timestep and Estimator-floor Audit

Numerical audit report

13 September 2026

1 Primary answer

The predeclared verdict is `MODERATE_AFFINITY_ASYMPTOTE_UNRESOLVED`. At the finest timestep, (6.5, 5.5) has $a_\infty/\Delta\beta = 0.993519$ with 95% CI [0.983582, 1.001209]. For (8, 4) the finest primary fit fails the frozen adequacy gate. The nominal equilibrium estimator floor is $1.088285\text{e-}04$ in slope units (0.0019 of $\Delta\beta = 0.057143$).

Every new case uses the same validated sampler, 128 independent streams, 31,252 duration-5 blocks per stream, and 4,000,256 raw rows. Both requested extra affinities were run at $dt = 2.5 \times 10^{-4}$ and 1.25×10^{-4} ; the $dt = 5 \times 10^{-4}$ trajectories are immutable baselines.

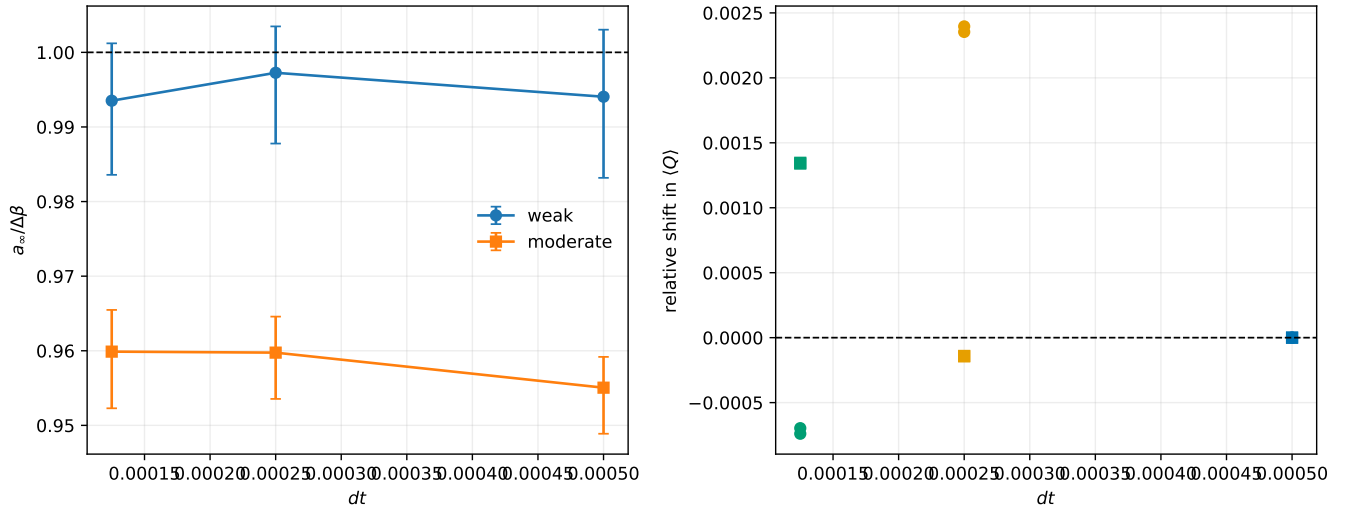


Figure 1: Left: finite-time intercept ratios at three timesteps. Right: relative mean-heat shifts at $\tau = 5$ and 25; colours encode timestep.

2 Finest-step three-affinity summary

affinity	selection	dt	$a_\infty/\Delta\beta$	95% CI	χ^2_ν	max $ r $	adequate	CI has 1	pass
(7, 5)	all_admissible	1.250e-04	0.995730	[0.988535, 1.000882]	0.645	1.461	yes	yes	yes
(8, 4)	all_admissible	1.250e-04	0.959884	[0.952289, 0.965475]	5.103	3.250	no	no	no
(6.5, 5.5)	all_admissible	1.250e-04	0.993519	[0.983582, 1.001209]	0.753	1.703	yes	yes	yes
(7, 5)	tau_ge_10	1.250e-04	1.000023	[0.990402, 1.006965]	0.109	0.528	yes	yes	yes
(8, 4)	tau_ge_10	1.250e-04	0.978279	[0.966215, 0.987892]	0.287	0.884	yes	no	no
(6.5, 5.5)	tau_ge_10	1.250e-04	0.994366	[0.982963, 1.003116]	0.863	1.718	yes	yes	yes

The direct finest-step criterion requires both an adequate $1/\tau$ model and a ratio CI containing one. An inadequate (8, 4) fit is reported as unresolved rather than converted into evidence against the fluctuation relation.

3 Per-window results at (6.5, 5.5)

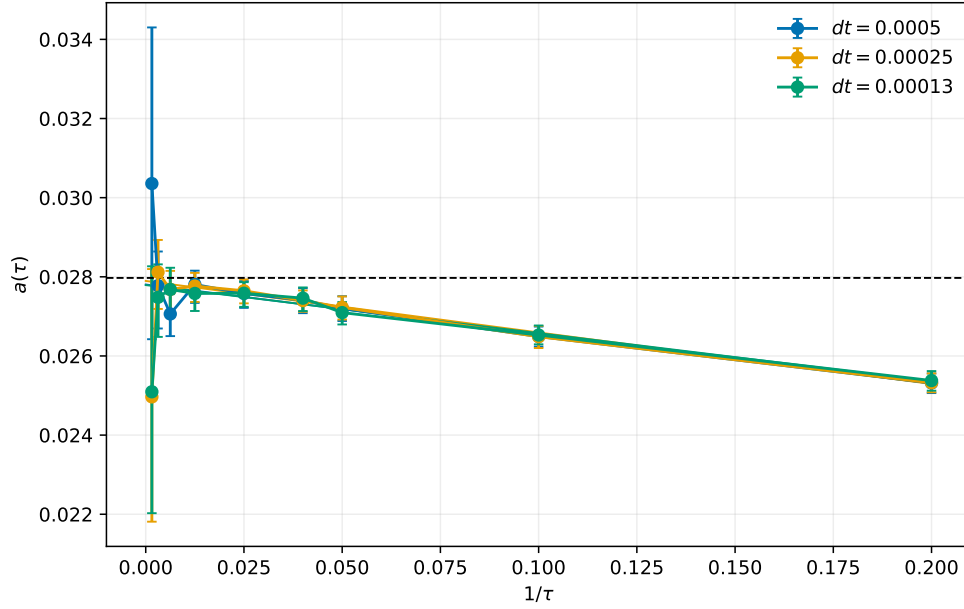


Figure 2: Weak-affinity finite-time slopes and frozen $1/\tau$ fits.

dt	τ	N	n_-	pairs	$a(\tau)$	bootstrap 95% CI	R^2	adm.
5.000e-04	5	4000256	1747665	99	0.025317	[0.025062, 0.025559]	0.9979	yes
5.000e-04	10	2000128	814568	84	0.026491	[0.026237, 0.026743]	0.9976	yes
5.000e-04	20	1000064	366513	74	0.027205	[0.026883, 0.027510]	0.9978	yes
5.000e-04	25	800000	280562	69	0.027407	[0.027081, 0.027712]	0.9976	yes
5.000e-04	40	499968	155995	65	0.027580	[0.027215, 0.027891]	0.9979	yes
5.000e-04	80	249984	60756	53	0.027806	[0.027341, 0.028155]	0.9979	yes
5.000e-04	160	124928	20136	43	0.027062	[0.026502, 0.027563]	0.9951	yes
5.000e-04	320	62464	5109	34	0.027776	[0.026694, 0.028640]	0.9944	yes
5.000e-04	640	31232	764	17	0.030356	[0.026423, 0.034302]	0.9704	yes
2.500e-04	5	4000256	1745814	99	0.025322	[0.025081, 0.025558]	0.9977	yes
2.500e-04	10	2000128	813275	86	0.026487	[0.026200, 0.026759]	0.9978	yes
2.500e-04	20	1000064	365872	74	0.027228	[0.026920, 0.027503]	0.9988	yes
2.500e-04	25	800000	279855	72	0.027400	[0.027117, 0.027655]	0.9977	yes
2.500e-04	40	499968	155701	65	0.027647	[0.027329, 0.027934]	0.9979	yes
2.500e-04	80	249984	60447	55	0.027766	[0.027365, 0.028100]	0.9980	yes
2.500e-04	160	124928	20012	44	0.027681	[0.027145, 0.028150]	0.9958	yes
2.500e-04	320	62464	5016	32	0.028116	[0.027188, 0.028932]	0.9940	yes
2.500e-04	640	31232	719	14	0.024966	[0.021812, 0.028197]	0.9510	yes
1.250e-04	5	4000256	1747636	98	0.025379	[0.025122, 0.025618]	0.9971	yes
1.250e-04	10	2000128	814370	85	0.026529	[0.026294, 0.026770]	0.9977	yes
1.250e-04	20	1000064	366079	76	0.027095	[0.026796, 0.027363]	0.9982	yes
1.250e-04	25	800000	280548	72	0.027461	[0.027135, 0.027732]	0.9985	yes
1.250e-04	40	499968	156076	63	0.027584	[0.027241, 0.027864]	0.9978	yes
1.250e-04	80	249984	60627	56	0.027579	[0.027136, 0.027943]	0.9974	yes
1.250e-04	160	124928	20138	44	0.027680	[0.027004, 0.028229]	0.9967	yes
1.250e-04	320	62464	5053	32	0.027482	[0.026482, 0.028315]	0.9926	yes
1.250e-04	640	31232	781	15	0.025095	[0.022027, 0.028266]	0.9640	yes

4 Per-window results at (8, 4)

dt	τ	N	n_-	pairs	$a(\tau)$	bootstrap 95% CI	R^2	adm.
5.000e-04	5	4000256	1078232	75	0.095942	[0.095619, 0.096242]	0.9997	yes
5.000e-04	10	2000128	359864	58	0.106818	[0.106332, 0.107204]	0.9997	yes
5.000e-04	20	1000064	88761	45	0.113369	[0.112612, 0.114019]	0.9995	yes
5.000e-04	25	800000	51218	37	0.116279	[0.115150, 0.117171]	0.9992	yes
5.000e-04	40	499968	12744	30	0.119661	[0.117436, 0.121338]	0.9975	yes
5.000e-04	80	249984	661	13	0.124603	[0.111797, 0.133624]	0.9866	yes
5.000e-04	160	124928	2	0	—	[—, —]	—	no
5.000e-04	320	62464	0	0	—	[—, —]	—	no

5.000e-04	640	31232	0	0	—	[—, —]	—	no
2.500e-04	5	4000256	1078192	75	0.095897	[0.095567, 0.096200]	0.9998	yes
2.500e-04	10	2000128	359207	58	0.106711	[0.106182, 0.107191]	0.9996	yes
2.500e-04	20	1000064	88994	44	0.114332	[0.113507, 0.115016]	0.9993	yes
2.500e-04	25	800000	51334	36	0.116946	[0.115858, 0.117787]	0.9993	yes
2.500e-04	40	499968	12748	31	0.119346	[0.117217, 0.121293]	0.9982	yes
2.500e-04	80	249984	646	14	0.112527	[0.103502, 0.123602]	0.9880	yes
2.500e-04	160	124928	2	0	—	[—, —]	—	no
2.500e-04	320	62464	0	0	—	[—, —]	—	no
2.500e-04	640	31232	0	0	—	[—, —]	—	no
1.250e-04	5	4000256	1077219	76	0.096213	[0.095882, 0.096497]	0.9997	yes
1.250e-04	10	2000128	359375	58	0.107221	[0.106663, 0.107665]	0.9996	yes
1.250e-04	20	1000064	88581	41	0.114660	[0.113662, 0.115405]	0.9995	yes
1.250e-04	25	800000	51323	38	0.116248	[0.115031, 0.117227]	0.9993	yes
1.250e-04	40	499968	12720	29	0.118734	[0.116338, 0.120576]	0.9986	yes
1.250e-04	80	249984	620	11	0.127176	[0.109144, 0.138208]	0.9862	yes
1.250e-04	160	124928	1	0	—	[—, —]	—	no
1.250e-04	320	62464	0	0	—	[—, —]	—	no
1.250e-04	640	31232	0	0	—	[—, —]	—	no

5 Finite-time extrapolations

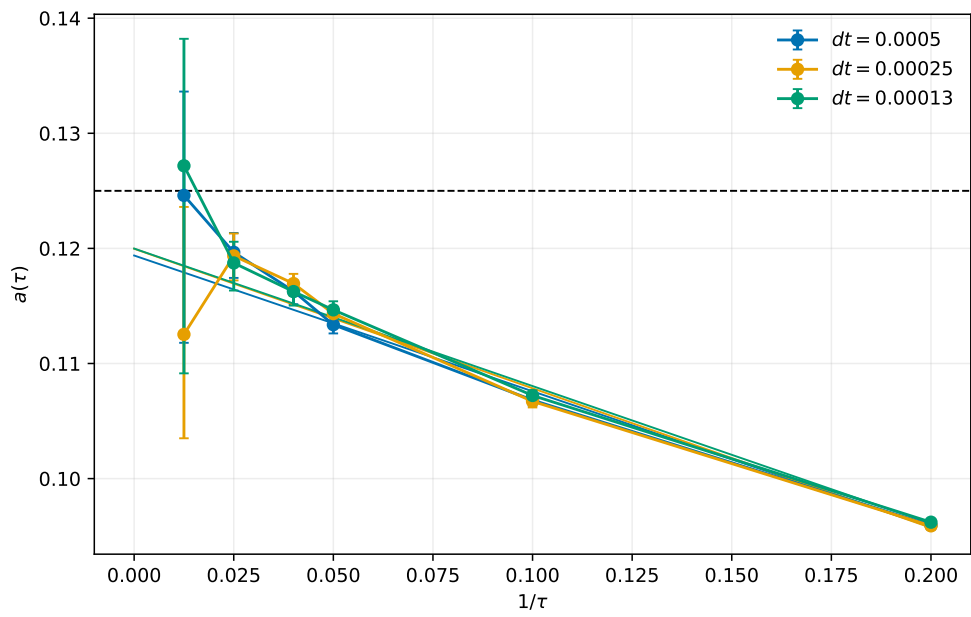


Figure 3: Moderate-affinity finite-time slopes and frozen $1/\tau$ fits.

(6.5, 5.5)

dt	selection	τ used	a_∞	95% CI	ratio	ratio CI	χ_ν^2	max $ r $	adequate
5.000e-04	all_admissible	5;10;20;25;40;80;160;320;640	0.027806	[0.027502, 0.028057]	0.994051	[0.983179, 1.003040]	1.265	2.380	yes
5.000e-04	tau_ge_10	10;20;25;40;80;160;320;640	0.027824	[0.027482, 0.028126]	0.994718	[0.982477, 1.005491]	1.466	2.437	yes
2.500e-04	all_admissible	5;10;20;25;40;80;160;320;640	0.027895	[0.027630, 0.028069]	0.997263	[0.987776, 1.003479]	0.700	1.833	yes
2.500e-04	tau_ge_10	10;20;25;40;80;160;320;640	0.027949	[0.027639, 0.028172]	0.999161	[0.988089, 1.007162]	0.732	1.865	yes
1.250e-04	all_admissible	5;10;20;25;40;80;160;320;640	0.027791	[0.027513, 0.028006]	0.993519	[0.983582, 1.001209]	0.753	1.703	yes
1.250e-04	tau_ge_10	10;20;25;40;80;160;320;640	0.027814	[0.027495, 0.028059]	0.994366	[0.982963, 1.003116]	0.863	1.718	yes

(8, 4)

dt	selection	τ used	a_∞	95% CI	ratio	ratio CI	χ_ν^2	max $ r $	adequate
5.000e-04	all_admissible	5;10;20;25;40;80	0.119381	[0.118610, 0.119897]	0.955051	[0.948878, 0.959180]	8.692	3.388	no
5.000e-04	tau_ge_10	10;20;25;40;80	0.121591	[0.120402, 0.122522]	0.972728	[0.963220, 0.980177]	3.466	2.175	no
2.500e-04	all_admissible	5;10;20;25;40;80	0.119969	[0.119193, 0.120572]	0.959750	[0.953546, 0.964579]	10.641	4.588	no
2.500e-04	tau_ge_10	10;20;25;40;80	0.122838	[0.121468, 0.123886]	0.982708	[0.971741, 0.991087]	1.776	1.622	yes
1.250e-04	all_admissible	5;10;20;25;40;80	0.119986	[0.119036, 0.120684]	0.959884	[0.952289, 0.965475]	5.103	3.250	no
1.250e-04	tau_ge_10	10;20;25;40;80	0.122285	[0.120777, 0.123487]	0.978279	[0.966215, 0.987892]	0.287	0.884	yes

$dt \rightarrow 0$ diagnostics

51

affinity	selection	r_0	95% CI	slope	slope CI	χ_ν^2	$O(dt)$	all adequate	same τ
weak	all_admissible	0.995263	[0.983447, 1.004505]	-0.280	[-33.610, 33.700]	0.445	yes	yes	yes
weak	tau_ge_10	0.996783	[0.982340, 1.007930]	-1.480	[-40.151, 42.524]	0.558	yes	yes	yes
moderate	all_admissible	0.962499	[0.953486, 0.969253]	-14.426	[-35.303, 6.386]	0.151	yes	no	yes
moderate	tau_ge_10	0.983983	[0.970072, 0.994762]	-20.410	[-55.558, 13.708]	1.032	yes	no	yes

The three-point $dt \rightarrow 0$ lines are diagnostics. The direct finest-step CI and the frozen adequacy gate take priority.

6 Mean-heat and slope shifts from $dt = 5 \times 10^{-4}$

(6.5, 5.5)

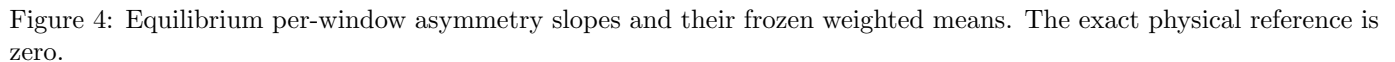
finer dt	τ	Δa	bootstrap 95% CI	contains 0	relative $\Delta\langle Q \rangle$
2.500e-04	5	4.783825e-06	[-3.606310e-04, 3.613549e-04]	yes	0.002396
2.500e-04	10	-3.485360e-06	[-3.787527e-04, 3.709913e-04]	yes	0.002396
2.500e-04	20	2.310966e-05	[-4.257795e-04, 4.677769e-04]	yes	0.002396
2.500e-04	25	-6.918930e-06	[-4.401585e-04, 4.225633e-04]	yes	0.002353
2.500e-04	40	6.743198e-05	[-3.897303e-04, 5.325325e-04]	yes	0.002339
2.500e-04	80	-3.945674e-05	[-6.062809e-04, 4.959351e-04]	yes	0.002339
2.500e-04	160	6.196482e-04	[-1.653264e-04, 0.001346]	yes	0.002329
2.500e-04	320	3.400612e-04	[-8.427722e-04, 0.001691]	yes	0.002329
2.500e-04	640	-0.005390	[-0.010549, -6.804849e-04]	no	0.002329
1.250e-04	5	6.179599e-05	[-2.942140e-04, 3.876489e-04]	yes	-6.966350e-04
1.250e-04	10	3.821577e-05	[-3.265993e-04, 3.627173e-04]	yes	-6.966350e-04
1.250e-04	20	-1.098043e-04	[-5.341226e-04, 3.025802e-04]	yes	-6.966350e-04
1.250e-04	25	5.387659e-05	[-4.164643e-04, 4.956783e-04]	yes	-7.395775e-04
1.250e-04	40	4.455938e-06	[-4.909261e-04, 4.539484e-04]	yes	-7.381693e-04
1.250e-04	80	-2.263072e-04	[-7.885918e-04, 3.928592e-04]	yes	-7.381693e-04
1.250e-04	160	6.186915e-04	[-2.568840e-04, 0.001388]	yes	-7.289794e-04
1.250e-04	320	-2.942128e-04	[-0.001559, 0.001047]	yes	-7.289794e-04
1.250e-04	640	-0.005261	[-0.010148, -2.100427e-04]	no	-7.289794e-04

(8, 4)

finer dt	τ	Δa	bootstrap 95% CI	contains 0	relative $\Delta\langle Q \rangle$
2.500e-04	5	-4.540900e-05	[-4.935718e-04, 3.693904e-04]	yes	-1.406187e-04
2.500e-04	10	-1.070960e-04	[-7.246587e-04, 5.377824e-04]	yes	-1.406187e-04
2.500e-04	20	9.630242e-04	[-9.530583e-05, 0.002074]	yes	-1.406187e-04
2.500e-04	25	6.664523e-04	[-7.351201e-04, 0.001974]	yes	-1.438952e-04
2.500e-04	40	-3.150634e-04	[-0.003101, 0.002796]	yes	-1.384996e-04
2.500e-04	80	-0.012076	[-0.025152, 0.005149]	yes	-1.384996e-04
2.500e-04	160	—	[-, -]	no	-1.076552e-04
2.500e-04	320	—	[-, -]	no	-1.076552e-04
2.500e-04	640	—	[-, -]	no	-1.076552e-04
1.250e-04	5	2.711069e-04	[-1.756673e-04, 7.119394e-04]	yes	0.001340
1.250e-04	10	4.033429e-04	[-2.107823e-04, 0.001042]	yes	0.001340
1.250e-04	20	0.001291	[1.124512e-04, 0.002344]	no	0.001340
1.250e-04	25	-3.138312e-05	[-0.001474, 0.001347]	yes	0.001346
1.250e-04	40	-9.271038e-04	[-0.003782, 0.002201]	yes	0.001347
1.250e-04	80	0.002573	[-0.016899, 0.018839]	yes	0.001347
1.250e-04	160	—	[-, -]	no	0.001344
1.250e-04	320	—	[-, -]	no	0.001344
1.250e-04	640	—	[-, -]	no	0.001344

7 Equilibrium estimator baseline

dt	τ	$a(\tau)$	bootstrap 95% CI	contains 0	n_-	pairs	R^2
5.000e-04	5	2.020979e-04	[-1.898629e-05, 4.288681e-04]	yes	2000032	106	0.0270
5.000e-04	10	1.393376e-04	[-1.190242e-04, 3.841128e-04]	yes	999544	97	0.0127
5.000e-04	20	1.630124e-04	[-9.526516e-05, 4.165654e-04]	yes	499786	85	0.0214
5.000e-04	25	4.721262e-05	[-1.819332e-04, 2.903563e-04]	yes	399455	81	0.0016
5.000e-04	40	-1.106779e-05	[-2.686678e-04, 2.622487e-04]	yes	249282	76	1.0247e-04
5.000e-04	80	3.895437e-05	[-2.037786e-04, 2.761613e-04]	yes	124666	71	0.0010
5.000e-04	160	2.266636e-04	[-4.544486e-05, 4.860217e-04]	yes	62425	66	0.0551
5.000e-04	320	-2.824073e-06	[-2.673825e-04, 2.463902e-04]	yes	31032	59	6.7076e-06
5.000e-04	640	1.206895e-04	[-1.337232e-04, 3.899440e-04]	yes	15544	54	0.0124
2.500e-04	5	-2.850192e-04	[-5.485820e-04, -2.332093e-05]	no	1999557	107	0.0510
2.500e-04	10	-1.775594e-04	[-4.344605e-04, 8.700647e-05]	yes	1000353	94	0.0222
2.500e-04	20	-4.324976e-05	[-3.053694e-04, 2.278426e-04]	yes	500592	83	0.0014
2.500e-04	25	-7.473530e-05	[-3.241626e-04, 1.906808e-04]	yes	400515	82	0.0037
2.500e-04	40	-8.034508e-05	[-3.324155e-04, 1.579096e-04]	yes	250368	78	0.0052
2.500e-04	80	-1.248807e-04	[-3.967967e-04, 1.291639e-04]	yes	125167	70	0.0125
2.500e-04	160	-1.375052e-04	[-3.994935e-04, 1.126236e-04]	yes	62616	65	0.0174
2.500e-04	320	-1.204334e-05	[-3.013355e-04, 2.491267e-04]	yes	31441	61	1.8736e-04
2.500e-04	640	-2.336626e-05	[-3.402746e-04, 2.227351e-04]	yes	15729	56	6.5743e-04



The fine-minus-coarse equilibrium weighted baseline is $-2.115232\text{e-}04$, with bootstrap SE $1.282241\text{e-}04$ and 95% CI $[-4.702495\text{e-}04, 2.809682\text{e-}05]$. The two baselines are statistically consistent under the frozen criterion. The nominal floor is $1.088285\text{e-}04$; its ratio to the residual (7, 5) finest-step deficit is 0.446, which is within the predeclared factor-three “same order” range.

308d843a15fee0cce1ea589da00f3d88155d3cfb5593981d85e81d328cfa0fa7.

case	kind	rows	SHA-256
weak_dt5e4	baseline	4000256	84ee7b05c008a142...
weak_dt2p5e4	new	4000256	c2c81dc939a8de7f...
weak_dt1p25e4	new	4000256	1b52519587878766...
moderate_dt5e4	baseline	4000256	336e934df7bd8555...
moderate_dt2p5e4	new	4000256	8a7df0b5ec26aba5...
moderate_dt1p25e4	new	4000256	315bf4a25af37e6d...
equilibrium_dt5e4	baseline	4000256	87af7c2eadcea396...
equilibrium_dt2p5e4	baseline	4000256	8f5024d0d4fa0f3f...

Exact commands and measured runtimes for the original runs are in `COMMANDS.tsv` and `RUNTIMES.tsv`; the two recovered fine runs are recorded separately in `RECOVERY_COMMANDS.tsv` and `RECOVERY_RUNTIMES.tsv`. The user-authorized power-policy override is documented in `POWER_POLICY_OVERRIDE.md`: it changed only when the already-running processes were allowed to execute, not any scientific setting. Complete per-window moments, raw symmetric-bin counts, all bootstrap draws, residuals, and hashes accompany this report. No scientific parameter, estimator rule, or acceptance gate was changed after production began.